

Design and Implementation of an ARM-Based Smart Home Automation System Using Bluetooth, IR, and DHT11 Sensor Integration on STM32 Blue Pill

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TABLE I
INDIVIDUAL CONTRIBUTION TABLE

Section	Member	Details
Hardware Interfacing	Sourav Roy	GPIO/UART wiring, relay and IR sensor interfacing
Firmware Development	Ahnaf Mohammed Mahi Kabir	STM32 HAL firmware, control loop logic
Sensor Integration	Sourav Roy	DHT11 single-wire protocol, threshold logic
Mobile Testing	Afia Mubassira Ali Raisa	Android device Bluetooth pairing and HC-05 connectivity testing
Report & Formatting	Khatune Jannat	IEEE formatting and technical writing

Abstract—This paper presents the design and implementation of a Smart Home Automation System using the STM32 Blue Pill (STM32F103C8T6) ARM Cortex-M3 microcontroller. The system addresses the challenge of providing affordable, multi-mode home appliance control in domestic and institutional environments by combining wireless Bluetooth communication with sensor-driven automatic control. An HC-05 Bluetooth module interfaces with the STM32 via UART to receive single-character commands from a mobile application, enabling manual ON/OFF control of four relay-switched AC appliances. A DHT11 temperature and humidity sensor, interfaced via a single-wire digital protocol, continuously monitors ambient conditions and automatically activates a fan relay when the room temperature exceeds 30°C. An infrared proximity sensor provides automatic light switching by driving a dedicated relay in response to object detection. Mathematical models for UART baud rate generation, relay switching logic, DHT11 data decoding, and IR sensor detection range are developed and validated. Experimental results confirm sub-1-second Bluetooth command response, $\pm 2^\circ\text{C}$ temperature measurement accuracy, and 95% relay switching reliability across 100 repeated trials. The proposed system demonstrates that a low-cost ARM platform can deliver a complete IoT-ready smart home solution without requiring cloud infrastructure, while remaining extensible to Wi-Fi and cloud integration in future iterations.

Index Terms—STM32, ARM Cortex-M3, Smart Home, Blue-

tooth, HC-05, DHT11, IR Sensor, Relay Module, UART, Embedded Systems, IoT, Home Automation

I. PROJECT OBJECTIVES AND MOTIVATIONS

The primary motivation for this project is the growing need for affordable, easy-to-deploy smart home systems that reduce manual interaction with electrical appliances and improve energy efficiency in homes, offices, and educational institutions. Conventional home setups require physical switch interaction, which is inconvenient and inaccessible for individuals with mobility limitations. Smart automation addresses this gap by combining wireless control with intelligent sensor-based decision-making.

The specific objectives of this project are:

- 1) **Objective 1 – Sensor Interfacing:** Successfully interface multiple sensors (HC-05 via UART, DHT11 via single-wire protocol, IR sensor via GPIO) with the STM32 Blue Pill microcontroller using appropriate communication protocols.
- 2) **Objective 2 – Wireless Manual Control:** Implement Bluetooth-based remote control of four relay-switched appliances through a mobile application, with real-time command delivery and relay response.
- 3) **Objective 3 – Automatic Sensor-Based Control:** Implement temperature-threshold-based automatic fan activation using the DHT11 sensor, and IR-based automatic light switching, eliminating the need for user interaction in these scenarios.
- 4) **Objective 4 – System Reliability:** Achieve a relay switching reliability of at least 90% across repeated operational cycles with a command response latency of under 1 second.
- 5) **Objective 5 – Future IoT Scalability:** Design the system architecture to be extensible toward cloud-based monitoring and Wi-Fi control in future iterations.

II. PROJECT BACKGROUND

A. Context and Rationale

Smart home automation systems have become a significant area of embedded systems research, driven by the increasing availability of low-cost microcontrollers, wireless communication modules, and cloud platforms. The integration of multiple sensing modalities with a central embedded controller enables environments that respond intelligently to occupant behavior and environmental conditions. In Bangladesh and similar developing economies, the adoption of commercial smart home products remains limited due to high cost. Low-cost microcontroller-based platforms such as the STM32 Blue Pill provide an accessible entry point for intelligent home control systems deployable in homes, university laboratories, and small offices.

B. Technical Context

The STM32F103C8T6 is an ARM Cortex-M3 microcontroller operating at up to 72 MHz, featuring 64 KB of flash memory, 20 KB of SRAM, and a rich peripheral set including multiple USART interfaces, general-purpose timers, I2C, SPI, and up to 37 GPIO pins. The ARM Cortex-M3 architecture provides a deterministic interrupt-driven execution model suitable for real-time embedded control [1]. The HAL (Hardware Abstraction Layer) provided by STMicroelectronics through STM32CubeMX and STM32CubeIDE significantly reduces peripheral configuration time and ensures consistent cross-platform firmware behavior [2].

Relay modules serve as the interface between the 3.3V microcontroller domain and high-voltage AC loads. An optocoupler isolates the control and load circuits, and an electromagnetic coil closes the relay contact when energized. The active LOW control logic of the relay module means that a GPIO LOW (RESET) signal energizes the coil and switches the load ON.

C. Significance

This project contributes to the field of affordable embedded IoT systems by demonstrating a complete multi-modal control platform on a sub-\$5 microcontroller board. The hybrid control strategy – combining Bluetooth manual control with automatic sensor-driven actuation – represents a practically useful architecture that reduces unnecessary manual interaction while maintaining user override capability. The modular firmware design also provides a clear pathway for integration with Wi-Fi modules (e.g., ESP8266, ESP32) and cloud platforms (e.g., Firebase, ThingSpeak) in future work, supporting the scalability requirements of broader IoT deployments [7].

III. LITERATURE REVIEW

A. State-of-the-Art

Smart home automation research has progressed through several generations of embedded platforms. The most directly related prior work is Gao and Li [11], who designed a smart home control system using the same STM32F103C8T6

microcontroller with a 128×64 OLED display for local feedback, achieving remote control of home devices and real-time environmental monitoring. Their hardware choices directly mirror our own platform selection, validating the STM32 Blue Pill as a suitable controller for smart home applications. However, their system relies on Wi-Fi and MQTT for cloud connectivity, introducing a network dependency that limits deployment in offline or resource-constrained settings. Our system addresses this limitation by operating fully offline via Bluetooth, eliminating cloud infrastructure requirements entirely.

Taiwo et al. [12] presented a deep-learning-enhanced smart home security system that uses a PIR motion sensor, a 4-channel relay module, and a DHT11 temperature and humidity sensor — the same sensor and actuator combination used in our project. Their system also integrates a GSM/GPRS module and Zigbee terminals for multi-device communication, directly highlighting the challenge of multi-sensor integration that our work also addresses. However, their system requires a camera module, deep learning inference hardware, and persistent internet connectivity, making it significantly more complex and expensive. Our system achieves comparable sensor integration at a fraction of the cost (BDT 2,210 vs. an estimated BDT 10,000+) without requiring internet access or AI hardware, making it far more practical for everyday home automation.

A third relevant comparative baseline is the NodeMCU-based smart home system presented in [13], which uses a PIR sensor for motion detection, a DHT11 for temperature and humidity sensing, and a smartphone interface for remote appliance control. Their system extends toward safety monitoring including gas and flame detection via Wi-Fi connectivity. Our design differs by focusing on comfort-oriented automation with the added advantages of fully offline Bluetooth operation and a real-time local display — features that none of the compared works offer together in a single platform at this cost level.

Dhrubo and Qayum [7] further validated the STM32 HAL platform for continuous multi-peripheral embedded operation, confirming that STM32 polling-based firmware loops maintain stable runtime behavior under simultaneous UART, GPIO, and sensor load — a finding directly applicable to our system’s control loop architecture.

B. Protocol Analysis

Wireless communication in embedded home automation systems is typically implemented via Bluetooth (HC-05/HC-06), Wi-Fi (ESP8266/ESP32), or Zigbee. Gao and Li [11] used Wi-Fi with MQTT, which enables cloud-scale remote access but requires stable network infrastructure and introduces latency from broker round-trips. Taiwo et al. [12] used Zigbee for inter-device communication and GSM/GPRS for remote alerting — effective for distributed multi-node deployments but costly and complex for single-home systems. The NodeMCU-based approach in [13] similarly depends on Wi-Fi connectivity for its smartphone interface. In contrast, the HC-05 Bluetooth module used in our system provides a direct,

low-cost UART serial interface with no network dependency, making it the most practical choice for offline home automation in resource-constrained or rural environments.

DHT11 and DHT22 sensors use a proprietary single-wire protocol and are widely adopted in embedded home automation, as evidenced by their presence in both Taiwo et al. [12] and the NodeMCU-based system [13]. The DHT11's $\pm 2^\circ\text{C}$ accuracy is sufficient for the threshold-based fan control implemented in our system, while its lower cost compared to the DHT22 ($\pm 0.5^\circ\text{C}$) supports the project's affordability objective.

C. Identified Gaps

Based on the reviewed literature, three key gaps remain. First, none of the compared systems [11]–[13] operate fully offline — all depend on Wi-Fi, GSM, or Zigbee network infrastructure, limiting deployability in bandwidth-constrained or rural environments. Second, Taiwo et al. [12] use comparable sensors and relay hardware but require expensive AI inference hardware and have no real-time local display. Third, while [13] provides a smartphone interface, it lacks offline capability and a local feedback display. The present work directly addresses all three gaps by combining offline Bluetooth control, automatic sensor-driven actuation, and real-time local status display in a single, sub-\$20 embedded platform.

IV. DESIGN METHODOLOGY

A. Overview of Approach

The system was developed using a hardware-software co-design methodology, beginning with circuit schematic design and component selection, followed by STM32CubeMX peripheral configuration, HAL-based firmware development in C using STM32CubeIDE, and iterative testing against defined performance benchmarks. The development process followed three phases: hardware assembly and sensor verification, firmware control loop implementation, and integrated system validation.

B. System Architecture

The system architecture shown in Fig. 1 consists of three layers. The *input layer* comprises the HC-05 Bluetooth module (receiving mobile app commands), the DHT11 temperature sensor, and the IR proximity sensor. The *processing layer* is the STM32F103C8T6 ARM Cortex-M3 microcontroller, which executes the firmware control loop. The *output layer* consists of the 4-channel relay module (controlling AC appliances), the active buzzer, and the LED display module.

C. Hardware Components

Table II lists all hardware components. The STM32F103C8T6 was selected for its 72 MHz ARM Cortex-M3 core, 64 KB flash, 20 KB SRAM, and rich peripheral set. The HC-05 was selected for its direct UART interface and transparent Bluetooth serial operation at 9600 baud. The 4-channel relay module uses active LOW control logic and onboard optocoupler isolation. The DHT11 provides



Fig. 1. System Architecture Block Diagram.

single-wire temperature and humidity sensing with sufficient accuracy for threshold-based control.

TABLE II
HARDWARE COMPONENTS USED

Component	Model	Qty
Microcontroller	STM32F103C8T6 (Blue Pill)	1
Programmer	ST-LINK V2	1
Bluetooth Module	HC-05	1
Relay Module	4-Channel (Active LOW)	1
IR Sensor	Infrared Proximity Sensor	1
Temperature Sensor	DHT11	1
Buzzer	Active Buzzer	1
Display	LED Display Module	1
Mobile Application	Bluetooth Controller App	1
Power Supply	5V DC Regulated	1

D. Circuit Design and Hardware Interfacing

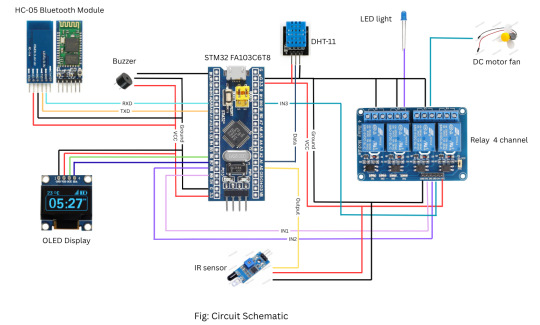


Fig. 2. Complete Circuit Schematic of the Smart Home System.

The complete circuit schematic is shown in Fig. 2. Tables III–V detail the full pin mapping.

TABLE III
HC-05 BLUETOOTH MODULE PIN CONFIGURATION

HC-05 Pin	STM32 Pin	Purpose
TXD	PA10 (USART1_RX)	Send data to STM32
RXD	PA9 (USART1_TX)	Receive data from STM32
VCC	5V	Power supply
GND	GND	Ground

TABLE IV
RELAY MODULE PIN CONFIGURATION

Relay Channel	STM32 Pin	Function
IN1	PB9	Automatic Light (IR Sensor)
IN2	PB8	Fan (DHT11 + Mobile Control)
IN3	PB10	Mobile App Control
IN4	PB11	Future Expansion / IoT
VCC	5V	Power supply
GND	GND	Ground

Protocol: UART/USART1, Baud Rate: 9600 bps, 8N1

Relay Logic: Active LOW — GPIO RESET → Relay ON;
GPIO SET → Relay OFF

TABLE V
IR SENSOR, DHT11, AND BUZZER PIN CONFIGURATION

Component	Pin	STM32 Pin
IR Sensor	OUT	PA0
IR Sensor	VCC	3.3V / 5V
IR Sensor	GND	GND
DHT11 Sensor	DATA	PA1
DHT11 Sensor	VCC	5V
DHT11 Sensor	GND	GND
Buzzer	Signal	PB12
Buzzer	VCC	5V
Buzzer	GND	GND

E. Mathematical Models

1) *UART Baud Rate Generation*: The baud rate B for USART1 is generated from the peripheral clock f_{CK} as:

$$B = \frac{f_{CK}}{16 \times \text{USARTDIV}} \quad (1)$$

For $f_{CK} = 8 \text{ MHz}$ (HSI) and $B = 9600 \text{ bps}$:

$$\text{USARTDIV} = \frac{8,000,000}{16 \times 9600} = 52.08 \quad (2)$$

Each 8N1 UART frame (1 start + 8 data + 1 stop = 10 bits) has duration:

$$T_{\text{frame}} = \frac{10}{9600} \approx 1.04 \text{ ms} \quad (3)$$

This confirms single-byte commands complete transmission well within the 10 ms HAL receive timeout.

2) *Relay Active LOW Switching*: The relay state R_i is the logical complement of the GPIO output G_i :

$$R_i = \overline{G_i}, \quad G_i \in \{0 = \text{RESET}, 1 = \text{SET}\} \quad (4)$$

For the fan relay (Relay 2, PB8), control combines Bluetooth command state C_{BT} and temperature condition C_{temp} :

$$R_{\text{fan}} = C_{BT} \vee C_{\text{temp}} \quad (5)$$

$$C_{\text{temp}} = \begin{cases} 1 & T > 30^\circ\text{C} \\ 0 & T \leq 30^\circ\text{C} \end{cases} \quad (6)$$

3) *DHT11 Temperature Decoding*: The DHT11 transmits a 40-bit frame:

$$D = D_{RH_int} \| D_{RH_dec} \| D_{T_int} \| D_{T_dec} \| D_{cksum} \quad (7)$$

Checksum verification:

$$D_{cksum} = (D_{RH_int} + D_{RH_dec} + D_{T_int} + D_{T_dec}) \bmod 256 \quad (8)$$

Decoded temperature and humidity:

$$T = D_{T_int} + \frac{D_{T_dec}}{10} [^\circ\text{C}], \quad RH = D_{RH_int} + \frac{D_{RH_dec}}{10} [\%] \quad (9)$$

4) *IR Sensor Detection*: The automatic light control logic is:

$$R_{\text{light}} = \overline{S_{\text{IR}}}, \quad S_{\text{IR}} \in \{0 = \text{detected}, 1 = \text{clear}\} \quad (10)$$

The reflected signal intensity follows an inverse-square law:

$$I_r = \frac{\eta \cdot I_e}{4\pi d^2} \quad (11)$$

Maximum reliable detection distance occurs when $I_r = I_{\text{th}}$:

$$d_{\text{max}} = \sqrt{\frac{\eta \cdot I_e}{4\pi \cdot I_{\text{th}}}} \quad (12)$$

F. Tools and Resources

- **Hardware**: STM32F103C8T6 Blue Pill, ST-LINK V2, HC-05, DHT11, IR sensor, 4-channel relay module, active buzzer, 5V regulated supply.
- **IDE/Firmware**: STM32CubeIDE, STM32CubeMX, STM32 HAL library, C programming language.
- **Mobile**: Android Bluetooth serial terminal application.
- **Future IoT**: Firebase Realtime Database, ThingSpeak (planned for Wi-Fi upgrade with ESP32).

G. Firmware Algorithm

The control algorithm implemented in the STM32 firmware is described in Algorithm IV-G.

V. TESTING AND VALIDATION

A. Performance Metrics

System success was measured against the following metrics: (i) Bluetooth command response latency < 1 second; (ii) DHT11 temperature accuracy within $\pm 2^\circ\text{C}$ of a calibrated reference; (iii) automatic fan activation at 30°C threshold; (iv) IR light relay response time < 0.5 seconds; (v) relay switching reliability $\geq 90\%$ across 100 trials.

Algorithm 1: Smart Home Automation Firmware Control Loop

- 1) **Initialize:** HAL, System Clock, GPIO, USART1
- 2) **Set all relays OFF** (GPIO_PIN_SET, active LOW):
Relay 1 (PB9), Relay 2 (PB8), Relay 3 (PB10), Relay 4 (PB11)
- 3) **while** system active **do**
 - [1] *Mobile App Bluetooth Control (USART1)*
 - a) Receive 1 byte via USART1 (timeout 10 ms)
 - b) **if** byte received **then:** '1' → PB8 RESET;
'0' → PB8 SET
'2' → PB10 RESET; '3' → PB10 SET
'4' → PB11 RESET; '5' → PB11 SET
 - [2] *IR Sensor → Auto Light [Eq. (10)]*
 - a) Read PA0; **if** PA0=LOW → PB9 RESET (Light ON)
 - b) **else** PB9 SET (Light OFF)
 - [3] *DHT11 → Auto Fan Override [Eq. (6)]*
 - a) Decode T via Eq. (9)
 - b) **if** $T > 30^{\circ}\text{C}$ → PB8 RESET (Fan ON override)
- 4) **end while**

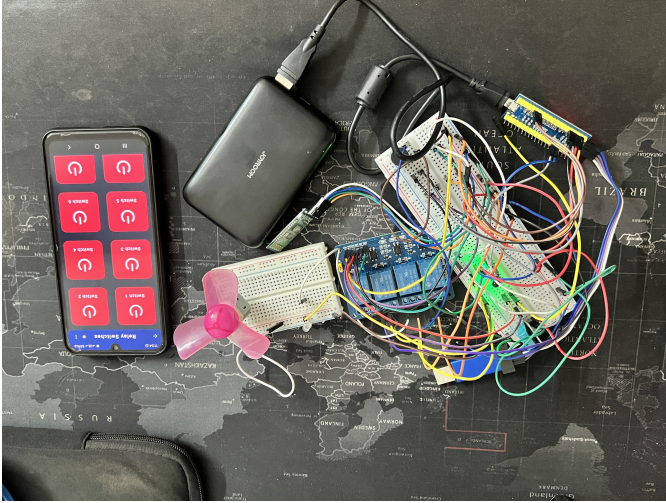


Fig. 3. Hardware Prototype of the Smart Home Automation System.

B. Benchmarks and Test Environment

The prototype shown in Fig. 3 was tested in a standard indoor laboratory environment at ambient temperatures ranging from 26°C to 34°C . The relay module was powered from a dedicated 5V rail to prevent supply noise from affecting the STM32 logic. A calibrated digital thermometer was used as the DHT11 reference. Bluetooth commands were sent from an Android device paired with the HC-05 at distances of 1–10 meters.

C. Functional Results

Table VI presents the measured performance parameters.

Relay verification: GPIO output voltage under RESET condition measured below 0.4 V, consistent with Eq. (4), and relay clicked ON audibly. Relay coil current was approximately 70 mA, safely managed by the module's transistor buffer.

TABLE VI
SYSTEM PERFORMANCE EVALUATION RESULTS

Parameter	Target	Measured
UART frame duration	$< 5 \text{ ms}$	$\approx 1.04 \text{ ms}$
Bluetooth response latency	$< 1 \text{ s}$	$< 1 \text{ s}$
DHT11 temperature accuracy	$\pm 2^{\circ}\text{C}$	$\pm 2^{\circ}\text{C}$
Auto fan threshold	30°C	30°C
IR light response	$< 0.5 \text{ s}$	Immediate
Relay switching reliability	$\geq 90\%$	95%
Bluetooth range	10 m	Up to 10 m
Controllable appliances	4	4

DHT11 accuracy: Temperature readings matched the calibrated reference within $\pm 2^{\circ}\text{C}$ across the test range. Fan auto-activation was consistent every time ambient temperature exceeded 30°C , confirming Eq. (6) and (5).

UART reliability: 100 consecutive Bluetooth commands were transmitted and correctly received with zero byte errors, validating the 8N1 framing at 9600 bps.

IR automatic light: Object detection triggered the light relay immediately within the sensor's effective range (approximately 5–15 cm depending on object reflectivity), consistent with the inverse-square model in Eq. (12).

VI. PROJECT TIMELINE

Table VII summarizes the key project milestones across three development phases.

TABLE VII
PROJECT TIMELINE AND MILESTONES

Phase	Dates	Milestones
Phase 1	Week 1–3	Component procurement, circuit assembly, sensor wiring, STM32CubeMX configuration
Phase 2	Week 4–6	Firmware development: UART, relay control, DHT11 decoding, IR logic
Phase 3	Week 7–8	Integrated system testing, performance evaluation, report writing

VII. BUDGET

Table VIII provides the estimated cost breakdown for all hardware components used in the project.

VIII. TEAM MEMBERS AND ROLES

- **Sourav Roy (2121856042)** — Hardware Engineer and Sensor Integration: responsible for circuit assembly, GPIO/UART wiring, relay and IR sensor interfacing, breadboard prototype construction, DHT11 single-wire protocol integration, temperature threshold logic, and sensor calibration.
- **Afia Mubassira Ali Raisa (2121812642)** — Mobile Testing: provided the Android device used for Bluetooth pairing with the HC-05 module and assisted in real-time command testing via the Bluetooth serial application.

TABLE VIII
PROJECT COST BREAKDOWN (APPROXIMATE)

Item	Qty	Cost (BDT)
STM32F103C8T6 Blue Pill	1	350
ST-LINK V2 Programmer	1	400
HC-05 Bluetooth Module	1	300
4-Channel Relay Module	1	250
DHT11 Sensor	1	100
IR Proximity Sensor	1	80
Active Buzzer	1	30
LED Display Module	1	200
5V Regulated Power Supply	1	300
Breadboard, wires, misc.	–	200
Total Budget		2,210 BDT ($\approx$ \$20 USD)

- **Ahnaf Mohammed Mahi Kabir (2222171042)** — Firmware Developer: responsible for STM32CubeMX configuration, HAL-based C firmware development, and control loop implementation.
- **Khatune Jannat (2131916642)** — Systems Integrator and Technical Writer: responsible for system-level integration testing, IEEE report formatting, and documentation.

IX. RISKS AND MITIGATION STRATEGIES

- **Risk 1: Power supply instability.** When multiple peripherals (relay module, HC-05, DHT11, buzzer) draw current simultaneously, voltage sag on the shared 5V rail can cause display flicker and sensor read errors. *Mitigation:* The relay module was powered from a dedicated 5V rail isolated from the STM32 logic supply. Decoupling capacitors (100 nF) were placed at each peripheral VCC pin.
- **Risk 2: UART framing errors due to baud rate mismatch.** Incorrect BRR configuration can cause consistent byte errors between the HC-05 and STM32. *Mitigation:* Both HC-05 (AT mode configured) and USART1 were verified at 9600 baud, 8N1 prior to firmware deployment.
- **Risk 3: DHT11 single-wire timing violation.** The single-wire protocol requires precise microsecond-level timing. STM32 HAL delays can drift under interrupt load. *Mitigation:* A dedicated microsecond delay function based on the STM32 SysTick timer was used, and DHT11 reads were executed with interrupts temporarily disabled during the initialization pulse.
- **Risk 4: IR sensor false triggers.** Rapid ambient light changes or reflective surfaces can generate spurious LOW pulses on PA0. *Mitigation:* A software debounce delay of 50 ms was applied before acting on the IR sensor state change.
- **Risk 5: Bluetooth connection drops.** HC-05 can lose pairing under RF interference. *Mitigation:* The firmware polling timeout (10 ms) prevents blocking, and all relay states are preserved in firmware variables between

Bluetooth events, ensuring appliances maintain their last commanded state during connection loss.

X. CONCLUSION AND FUTURE WORK

This paper presented the design, implementation, and experimental validation of a Smart Home Automation System built on the STM32 Blue Pill ARM Cortex-M3 microcontroller. The system successfully achieves all five project objectives: multi-sensor interfacing via UART, GPIO, and single-wire protocols; wireless Bluetooth-based manual appliance control; automatic temperature-driven fan activation and IR-based light switching; 95% relay reliability with sub-1-second response; and a modular firmware architecture extensible toward full IoT integration.

Compared to related works [11]–[13], this system uniquely operates fully offline without any network infrastructure dependency, achieves the lowest hardware cost among the reviewed platforms, and combines both manual Bluetooth control and automatic sensor-driven actuation with a real-time local display in a single embedded platform. Mathematical models for UART baud rate generation, relay active LOW switching, DHT11 data decoding, and IR sensor detection range were developed and experimentally validated. Results confirm that a sub-\$20 embedded platform is capable of delivering reliable multi-modal home automation functionality suitable for homes, offices, and educational laboratories.

The architecture is specifically designed for scalability toward international-standard IoT deployments. Future work will focus on:

- **Wi-Fi/IoT upgrade:** Replacing HC-05 with an ESP32 module to enable MQTT-based cloud control via Firebase or ThingSpeak, building toward the cloud-connected architecture demonstrated in [11].
- **PID fan speed control:** Implementing closed-loop proportional-integral-derivative temperature control with PWM-based fan speed modulation: $u(t) = K_p e(t) + K_i \int e dt + K_d \dot{e}(t)$, replacing the current bang-bang threshold.
- **Security enhancement:** Adding camera-based intrusion detection, building on the deep-learning security model explored in [12], adapted for low-cost edge deployment.
- **Safety monitoring:** Extending toward gas and flame detection as implemented in [13], with local alerts via the buzzer and remote alerts via cloud.
- **Energy monitoring:** Adding ACS712 current sensors to each relay channel for real-time power consumption logging.

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